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TWO-WHEEL INVERTED PENDULUM ROBOT: DESIGN, MODELING, SIMULATION AND CONTROL

Edo Walfrido de Almeida
Guilherme Oliveira Braz
Gabriel Ribeiro Tartari
José Matheus Santos
Leonardo Murilo Nepomuceno
Helon Vicente Hultmann Ayala
Pontifical Catholic University of Paraná (PUCPR) - Rua Imaculada Conceição, 1155 – Prado Velho, Curitiba - PR, 80215-901.
edo.almeida, guilherme.braz, gabriel.tartari, j.santos15, leonardo.murilo, helon.ayala@pucpr.br

Abstract. *This paper presents the development of a functional two-wheeled inverted pendulum prototype, with an emphasis on its mechanical assembly, electronic integration, and the implementation of low-level control for the DC motors. The system was designed with an educational and experimental focus, utilizing accessible components and widely available tools, such as programmable microcontrollers and position sensors. Initially, the mechanical structure of the platform and the embedded control circuitry are described, followed by the selection of actuators and the implementation of the speed control algorithm. The dynamic response of the motors was experimentally characterized through step input tests, enabling the identification of parameters for the design of digital controllers. The results demonstrate the viability of the proposed architecture as a foundation for further studies in the control of unstable systems. This work aims to contribute a practical and reproducible approach for didactic and experimental applications in the fields of robotics and control engineering.*

Keywords: *two-wheeled inverted pendulum, low-level control system, DC motor, digital control, educational robotics*

1. INTRODUCTION

The control of unstable systems is a classic topic in control engineering, with applications in robotics, autonomous vehicles, and embedded systems. A representative case is the Two-Wheeled Inverted Pendulum (TWIP), which requires real-time strategies to maintain vertical balance. Due to its nonlinear and highly coupled dynamics, the TWIP is widely used as a didactic model for validating control techniques (Zhuang *et al.*, 2014).

Besides its pedagogical use, the TWIP is conceptually related to self-balancing robots like the Segway, applying inverted pendulum principles to achieve stable locomotion (Grasser *et al.*, 2002). Its dynamic resemblance to humanoid robots makes it suitable for studies on balance and motion. Its simple construction enables practical implementation of both classical and modern controllers, such as PID, LQR, and adaptive control.

This work proposes the development of a TWIP-based prototype for validating control strategies and analyzing system dynamics. The structure uses 3D-printed components, locomotion motors, and space for future servomotor integration. The electronics are currently tested on a protoboard and will be embedded later, a typical approach in mobile robotics and real-time control research (Lee and Jung, 2012). The TWIP is widely studied in the literature for its combination of educational relevance and application potential in mobile robotics (Klemm *et al.*, 2020) (Lin and Tsai, 2009).

In practical applications, the TWIP serves as a conceptual basis for robots like Ascento, where balance control is essential for locomotion (Klemm *et al.*, 2019). Recent studies expand these systems to more advanced mobile robots, integrating inertial sensors and microcontrollers for real-time control, aiming at intelligent mobility and autonomous robotics (Grandia *et al.*, 2024; Christen *et al.*, 2025). Nevertheless, challenges such as nonlinearities, postural variations, and unknown equilibrium states demand robust control approaches (Liu *et al.*, 2024). Low-cost prototypes using 3D-printed parts and programmable microcontrollers have been explored to facilitate experimentation, though there is still a gap in detailed documentation of low-level control implementations at early development stages (Brentari *et al.*, 2015).

In this context, this work presents a functional prototype built with modular architecture and accessible components, aiming to validate classical techniques such as PI and pole placement directly on hardware. The proposal contributes a systematic and reproducible study, useful for educational purposes and as a foundation for developing embedded stabilization algorithms.

2. PROTOTYPE DEVELOPMENT

The two-wheeled inverted pendulum (TWIP) prototype was developed to validate the feasibility of an embedded control system, focusing on the mechanical structure, electronic integration, and initial testing. This section presents the assembly of the experimental setup, covering both the physical structure and software implementation[1], and establishes a modular foundation for future system expansions.

2.1 Mechanical Structure

The TWIP prototype structure was originally conceived as a lightweight, robust, and adaptable mechanical base, and made publicly available online (Skythinker616, 2023). In the present work, we adapt that design to accommodate off-the-shelf components, replacing certain custom parts proposed by the authors. Our contribution thus builds on (Skythinker616, 2023), tailoring the prototype to rely on standard components that are readily available in the Brazilian market. Built with 3D-printed PLA components, the platform enables easy customization and quick part replacement. The chassis accommodates two symmetrically positioned wheels, ensuring balanced weight distribution and improved stability during initial testing.

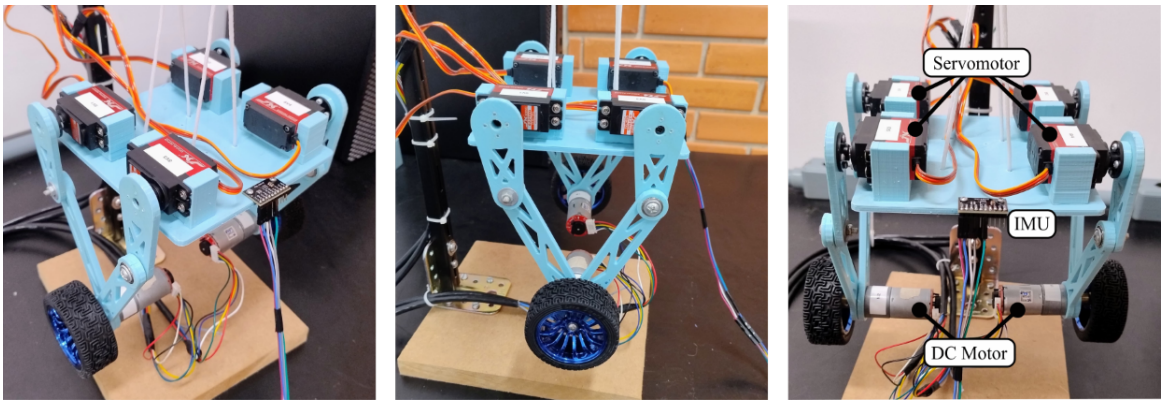

Figure 1. Overview (left and center) and main components (right) of the robot prototype on the test bench.

The DC motors were mounted on 3D-printed lateral supports featuring a triangular geometry to enhance rigidity and with strategically placed holes to ensure firm anchoring, providing mechanical stability and protection. For future system expansion, additional supports with passive bearing-based joints were incorporated, currently locked in this initial phase. Auxiliary servomotors were also pre-installed on the chassis to assist in stability and tilt adjustment when overcoming obstacles in later stages, although they are not yet integrated into the control system at this initial stage.

Table 1. Structural components used in the TWIP prototype

Component	Description
Printing material	PLA (Polylactic Acid), chosen for its lightness, low cost, and ease of part replacement
Central chassis	3D-printed structure with mounts for microcontroller, motors, and servomotors
Motor mounts	Side parts printed in 3D, with holes for DC motor attachment
Lateral bearings	Used in the passive joints of the robot's "legs"; fixed in this initial phase
Screws	Self-tapping screws used in the general assembly of the structure
Top platform	Flat base designed for installing sensors, servos, or electronic modules

The chassis was assembled using self-tapping screws, ensuring firm fixation and flexibility for adjustments during development. This modular assembly provides a solid foundation for future integration and validation of locomotion and control systems.

2.2 Electronic Circuit

The electronic circuit of the TWIP prototype was designed to control the DC motors and process data in real time. It is currently in the bench-testing phase, assembled on a protoboard. The system is based on the ESP32 microcontroller, selected for its processing capacity, wireless connectivity (Wi-Fi/Bluetooth), and support for real-time operations.

[1]The source code, hardware design files, and experimental data related to this study are publicly available at https://github.com/EdoAlmeida/TWIP-Robot.

Motor control is handled by an L298N driver board, which actuates two DC motors with incremental encoders for speed and position measurement, enabling closed-loop control. Power is currently supplied by a 12V external source, with plans to replace it with rechargeable 6V batteries to increase mobility and integration into the robot.

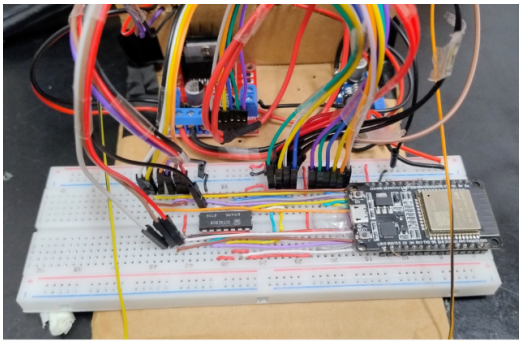

Figure 2. Electronic components assembled on a protoboard, positioned externally to the main structure of the prototype.

Connections between components use jumper wires, providing flexibility during testing. The integration of an inertial measurement unit (IMU) is planned to provide tilt data and support stability control.

Table 2. Main electronic components used in the TWIP prototype

Qtd.	Componente	Descrição
4	6221MG Servo Motor	Digital micro servo with metal gear, used for tilt adjustment and auxiliary joints
2	JGA25-371 DC Motor	6V DC motor, 210 RPM, with output shaft and integrated encoder for speed measurement
1	ESP32-WROOM-32	Microcontroller with integrated Wi-Fi and Bluetooth, dual-core
1	ICM-20948 IMU	Inertial measurement unit with 6-axis accelerometer and gyroscope for tilt and orientation sensing
1	L298N H-Bridge Module	Power driver for bidirectional control of two DC motors
1	12V Switching Power Supply	Power supply with stabilized 12V DC output
1	Protoboard	Breadboard for temporary electronic circuit assembly
1	74HC08 IC	Integrated circuit with four AND gates, CMOS technology
1	LM2596 Voltage Regulator	Adjustable step-down buck converter

Current tests focus on the open-loop characterization of the DC motors, which has been completed, and the initial validation of closed-loop control, with results presented in this paper. The circuit operates independently from the mechanical structure, facilitating module verification and optimization of communication between the ESP32, L298N, and encoders before integration into the prototype.

2.3 Actuator Selection

The actuators for the TWIP prototype were empirically selected based on initial torque and speed requirements to enable dynamic balance and locomotion, with performance validation planned throughout the project development.

For the wheel drive, two JGA25-371 DC motors were chosen. These motors operate at 6 V, compatible with the projected battery system, and offer a no-load speed of 360 RPM and torque ranging from 0.5 to 1.1 kg·cm. Their characteristics suit the lightweight PLA-printed chassis and enable quick response for stabilization control. Integrated encoders allow accurate velocity and position measurements, essential for closed-loop control implementation.

To support future adjustments in tilt and stability—particularly in uneven terrain—6221-MG metal-geared servomotors were pre-installed. Operating at 6 V, these servos provide 6 kg·cm of torque and 0.16 s/60° speed, enabling precise and fast inclination corrections. Both actuators were selected for their compatibility with the system's scale and power supply, as well as for their market availability, ensuring an accessible and functional starting point for experimental validation.

2.4 Development with Software Tools

The TWIP prototype development employed specific tools for programming, data acquisition, and simulation, which were essential for the initial validation of the control system. The ESP32 was programmed using the Arduino IDE,

selected for its accessibility, support for compatible libraries, and active developer community, which facilitated the implementation of control algorithms and initial testing of DC motors and encoders.

Experimental data were collected using PLX-DAQ, an Excel add-in that records variables via serial communication with the ESP32. This setup enabled preliminary open- and closed-loop analysis based on real-time data.

Dynamic simulations were performed in Matlab/Simulink using models derived from the system's physical equations to predict its response and support the definition of initial control parameters.

3. CONTROLLER DESIGN

Aligned with the purpose of this article—to offer a practical and reproducible guide for controlling unstable systems—the controller design adopts a systematic and incremental approach. The initial focus is on low-level speed control of the DC motors, which is essential to ensure predictable and stable actuator response. This provides a robust foundation for the later implementation of balance algorithms.

The methodology employed in this section consists of four main steps: 1) experimental characterization of the motors to obtain an accurate mathematical model; 2) design and optimization of a digital controller in a simulation environment; 3) implementation of the control algorithm on the ESP32 microcontroller; and 4) comparative validation between simulated results and experimental data from the physical prototype.

3.1 Modeling and Experimental Characterization

Before designing the speed controller, it is essential to understand the full dynamics of the TWIP system. The following presents the derivation of the equations of motion using the Lagrangian formulation, which describes the interactions between the base position and the pendulum angle.

The generalized coordinates, illustrated in Figure 3, describing the system configuration are q_1, horizontal position of the base (mass m); q_2, the pendulum angle with respect to the vertical (mass M). The total kinetic energy of the system (T) is the sum of the energies of the base (m) and the suspended mass (M), while the potential energy (V) arises from the height of mass M. Their expressions are:

$$T = \frac{1}{2}(m+M)\dot{q}_1^2 + ML\dot{q}_1\dot{q}_2\cos(q_2) + \frac{1}{2}ML^2\dot{q}_2^2, \tag{1}$$

$$V = MgL\cos(q_2), \tag{2}$$

where L is the length of the pendulum rod and g is the gravitational acceleration.

Using Lagrangian mechanics, the equations of motion are obtained as below.

$$(m+M)\ddot{q}_1 + ML\ddot{q}_2\cos(q_2) - ML\dot{q}_2^2\sin(q_2) = f \tag{3}$$

$$ML^2\ddot{q}_2 + ML\ddot{q}_1\cos(q_2) - MgL\sin(q_2) = 0 \tag{4}$$

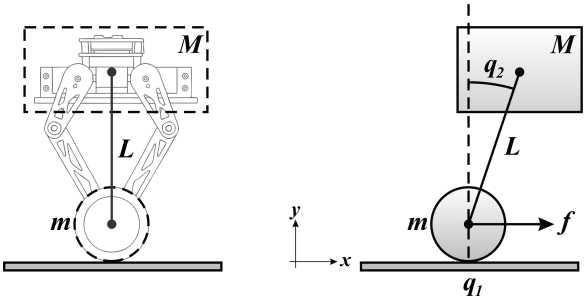

Figure 3. Schematic of the TWIP robot (left) and its free-body diagram (right), illustrating the generalized coordinates: base position (q_1) and pendulum angle (q_2).

3.1.1 Step Response and Model Simplification

A step input signal was applied to the motors, varying the voltage from 0V to 6V (equivalent to a PWM value of 255), using the L298N driver board. The angular speed of each motor was measured in RPM via the built-in encoders, and the

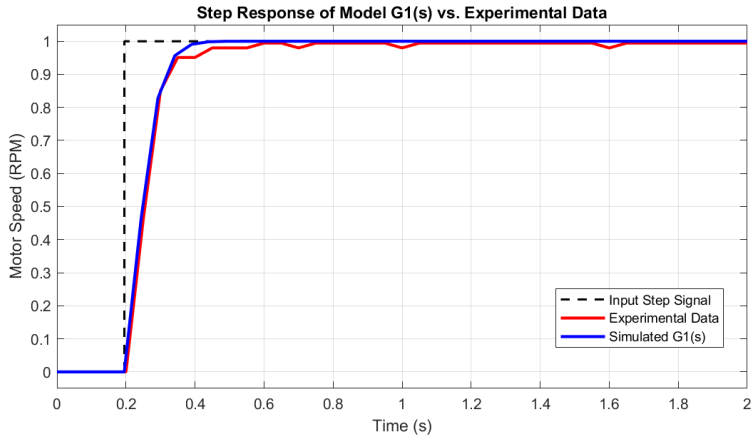

Figure 4. Step input of PWM and open-loop motor speed (RPM) responses.

time, PWM signal, and motor speeds were recorded in real-time into an Excel spreadsheet using PLX-DAQ, as described in section *Development with Software Tools*, and later imported into MATLAB. Figure 4 depicts the measured data and the simulated model fitted to it, detailed next.

With the experimental data in MATLAB, the identification for each motor followed four main steps: data preparation (defining time, input, and output vectors, with offset removal), creation of the $iddata$ object with the sampling period, estimation of the continuous transfer function using $tfest$ with a two-pole model, and model validation by comparing the simulated response with the experimental data. After analysing the dominant poles and correcting for the transfer function gain, for each motor we found the following models

$$G_1(s) = \frac{9.597}{0.106s + 1}, \tag{5}$$

$$G_2(s) = \frac{9.548}{0.1065s + 1}, \tag{6}$$

which are quite similar, though not equal, transfer functions for each motor that are slightly different.

These first-order transfer functions, now justified by the identification and simplification process, form the basis for the digital controller design, which will be detailed in the next section.

3.2 Controller Design

With the transfer functions of the JGA25-371 DC motors properly characterized and the first-order linear models $G_1(s)$ and $G_2(s)$ obtained, the next step is designing a closed-loop speed controller. The main goal of this controller is to ensure the motor speed accurately and stably follows a desired reference, with an appropriate transient response. A key performance requirement is zero overshoot to a step reference, combined with optimized settling time.

As shown by the approximate transfer functions $G_1(s)$(5) and $G_2(s)$(6), both motors exhibit very similar gain and time constant parameters. Therefore, to simplify and streamline the design process, the speed controller will be tuned using the model for Motor 1, $G_{\text{M1}}(s) = G_1(s)$. It is assumed that the controller designed this way will deliver comparable performance when applied to Motor 2, a hypothesis that will be experimentally verified in later stages of this work.

3.2.1 Control Strategy and Continuous-Time Tuning

For the motor speed control, a specific controller was designed with transfer function $C(s) = 0.85\frac{s+8.5}{s+0.2}$ was designed via MATLAB's PID Tuning toolbox combined with pole placement. The tuning objective was to ensure a closed-loop step response with zero overshoot and minimal settling time while preserving system stability. The controller parameters were optimized to place closed-loop poles at desired locations on the complex plane, achieving the specified transient dynamics. Figure 5 presents the block diagram of the implemented continuous-time closed-loop speed control system.

Controller performance was validated by MATLAB/Simulink simulation. The controller was discretized to obtain a difference equation and implemented in C++ using a Zero-Order Hold (ZOH) discretization method in order to run it in real-time, operating at 100 Hz.

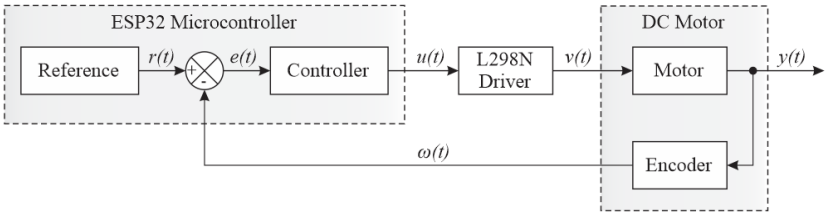

Figure 5. Block diagram of the continuous closed-loop speed control system.

3.3 Implementation on the ESP32

With the design and validation of the digital controller completed in simulation, the next step consisted of its practical implementation on the ESP32-WROOM-32 microcontroller. The code was developed in the Arduino IDE environment, using libraries for encoder reading and interfacing with motor drivers.

The real-time implementation of the control algorithm was structured into three parts: speed measurement, periodic execution of the control loop, and actuation.

Speed measurement. The motor speed ($y[k]$), used as feedback, was estimated from encoder counts using the `ESP32Encoder` library. At each control cycle, the current encoder count (`atual`) is compared to the previous value (`ultimaLeitura`). The difference in counts (Δn) is scaled by the sampling interval (T_s), expressed in milliseconds, to obtain pulses per second (PPS) and then converted to revolutions per minute (RPM) according to

$$rpm = \frac{60 \times (1000 \times \Delta n)}{T_s \times \text{PPR}} \tag{7}$$

where the factor 1000 converts the sampling interval from milliseconds to seconds, and *PPR* denotes the number of pulses per revolution of the encoder.

Timed execution. To ensure a constant sampling period of $T_s = 10\,\text{ms}$, the control loop was executed periodically using a non-blocking structure based on the `millis()` function. At each iteration, the elapsed time since the last control update is compared with the defined sampling interval. When the elapsed time exceeds T_s, the controller is executed, and the reference time is updated. This approach guarantees precise timing for the control task without interrupting other processes running on the ESP32.

Control and actuation. The discrete controller was implemented through a difference equation using the previously obtained coefficients and the past values of both the error and the control signal. At each sampling instant, the current speed error is calculated as the difference between the reference value and the measured speed. The corresponding update procedure for the control signal is presented in Figure 6, which illustrates the implementation structure used in the ESP32 code.

```
// ---Discretized Controller Parameters (obtained from MATLAB) ---
const float b0_controller = 0.8500;
const float b1_controller = -0.7778;
const float a1_controller = -0.9980;

float current_error = currentTargetRPM - abs(rpm_atual);
float u_k = (-a1_controller * u_prev) + (b0_controller * current_error) + (b1_controller * error_prev);
error_prev = current_error;
u_prev = u_k;
motorPWM_output = constrain(round(u_k), 0, 255);
```

Figure 6. Excerpt of the ESP32 code implementing the discrete difference equation for control.

The resulting control signal (u_k) was limited using the `constrain()` function to ensure that its value remained within the valid PWM range (0–255). This saturation step prevents overdriving the motors while maintaining the actuator within its safe operating region.

3.4 Experimental Results Analysis

The presented plots correspond to the experimental validation tests of the speed controller implemented on the ESP32. Two main tests were conducted: one with a stepwise reference signal and another with a sinusoidal reference, in order to evaluate the system's tracking performance under different scenarios.

It is important to note that, although the control loop operates with a sampling period (T_s) of 10 ms, the data acquisition for visualization was performed every 50 ms. This lower sampling rate reflects the performance limit of the PLX-DAQ

software to record real-time data without loss. As a result, the graphical representation of the measured data may appear more "jagged" or lower in resolution than the actual system behavior, especially during rapid transitions.

3.4.1 Test 1: Response to Sequential Step Inputs

The performance of the closed-loop controller was evaluated through a test involving a sequence of speed step references. Figure 7 presents the results, comparing the desired reference trajectory with the actual motor response and the corresponding control signal behavior. The measured output ("RPM Act", in red) demonstrates excellent reference-

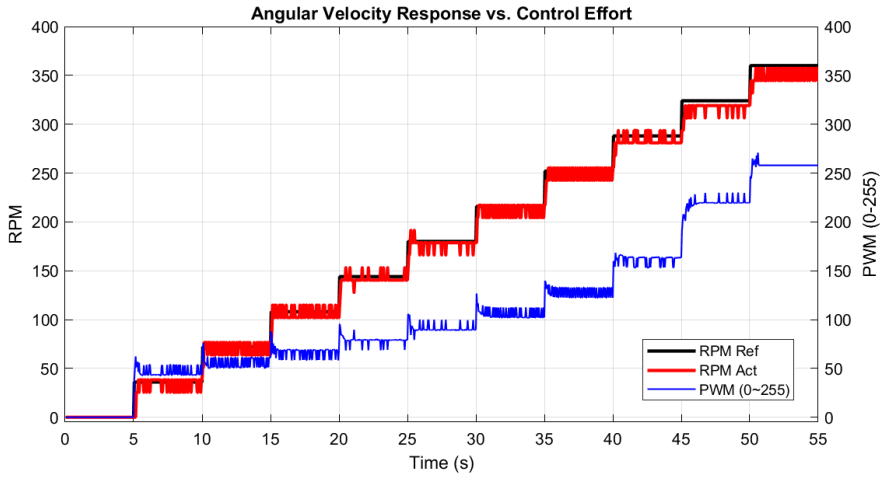

Figure 7. Speed control validation. The plot compares the motor's response (RPM Measured) with the step reference (RPM Reference). The control signal (PWM Feedback) is also included to assess the system's actuation effort.

tracking capability when compared to the step input ("RPM Ref", in black). The system responds quickly to each new setpoint, stabilizing at the new speed level with precision and no apparent overshoot, thus validating the controller design and tuning. The small oscillations observed at each level are expected in real systems due to measurement noise and mechanical or electrical nonlinearities.

The PWM control signal ("PWM (0-255)", in blue) behaves as expected for this control strategy. At each step change, the controller applies a brief peak in PWM to accelerate the motor rapidly, followed by a steady-state value sufficient to maintain the target speed. It is also noticeable that, at higher reference values, the PWM signal approaches its saturation limit, indicating that the controller is operating near its maximum actuation capacity.

3.4.2 Test 2: Response to a Sinusoidal Reference

The controller's dynamic performance was evaluated using a sinusoidal reference trajectory, which represents a more demanding test than step inputs due to the continuous variation in speed and acceleration. Figure 8 presents the results using a dual Y-axis plot, enabling a clear visualization of both the system's output speed and the corresponding control effort.

The measured output speed ("RPM Act", red curve) shows excellent tracking performance relative to the sinusoidal reference ("RPM Ref", black curve). The controller maintains the speed trajectory with high fidelity, exhibiting only a small phase lag and slight attenuation near the peaks—typical and acceptable behaviors for dynamic tracking tasks. These results confirm the controller's ability to follow continuously varying commands with precision and stability.

The control signal ("PWM (0-255)", blue curve) shown on the right Y-axis reflects the actuation effort required to achieve this performance. The PWM waveform resembles a distorted sinusoid, shaped by the controller's attempts to compensate for nonlinearities and system inertia. This distortion is expected in practical applications, where the control signal must adapt continuously to match the desired output. The dual-axis visualization effectively highlights the correlation between the controller's actuation and the resulting motor response, reinforcing the system's dynamic tracking capability.

3.4.3 Conclusion of the Analysis

The experimental results successfully validate the controller design. The system demonstrated the ability to follow both step and dynamic (sinusoidal) references with high accuracy, meeting the requirement of zero overshoot. The analysis of the PWM control signal provides valuable insights into saturation and system noise, paving the way for future optimizations.

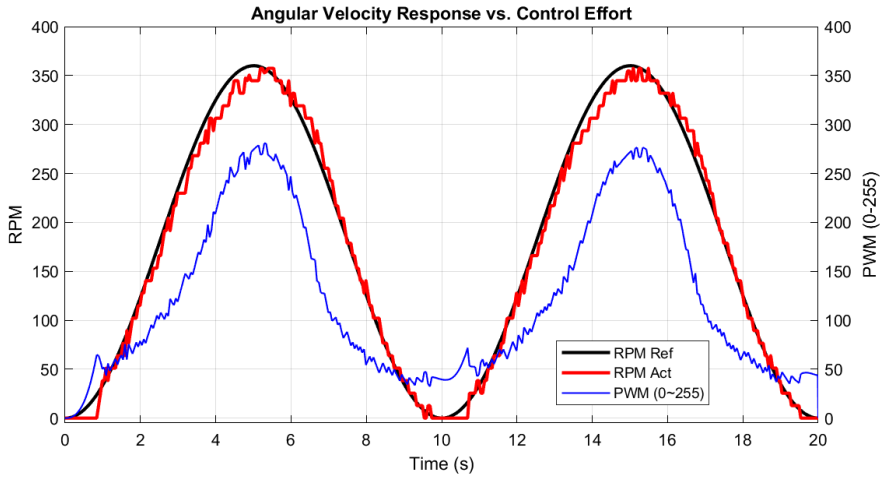

Figure 8. Closed-loop response to a sinusoidal trajectory. The left axis is motor speed (RPM); the right axis is the control signal (PWM). This dual-axis plot enables a direct comparison between system output and control effort.

4. EXPERIMENTAL CHARACTERIZATION OF THE INERTIAL MEASUREMENT UNIT (IMU)

Accurate and fast measurement of the chassis angular orientation is the most critical requirement for the balance control of a TWIP robot. For this purpose, the ICM-20948 Inertial Measurement Unit (IMU) was selected. This sensor stands out for integrating a three-axis accelerometer, gyroscope, and magnetometer, in addition to an embedded Digital Motion Processor (DMP). The DMP executes 9-axis sensor fusion algorithms directly on the hardware, providing pre-computed Euler angles (Pitch, Roll, Yaw). This feature is advantageous, as it significantly reduces the computational load on the main microcontroller (ESP32), allowing it to focus on high-level control tasks (InvenSense, 2024).

To ensure the reliability of the orientation data that will feed the control system, a rigorous experimental characterization of the IMU was carried out. The process was divided into three sequential stages to validate the sensor under static, controlled dynamic, and adverse dynamic conditions.

4.1 Test 1: Static Analysis (Baseline)

The first test aimed to evaluate the sensor performance under static conditions, establishing a baseline for its accuracy and stability. The IMU was placed on a flat, leveled surface, and its data were recorded for 30 seconds.

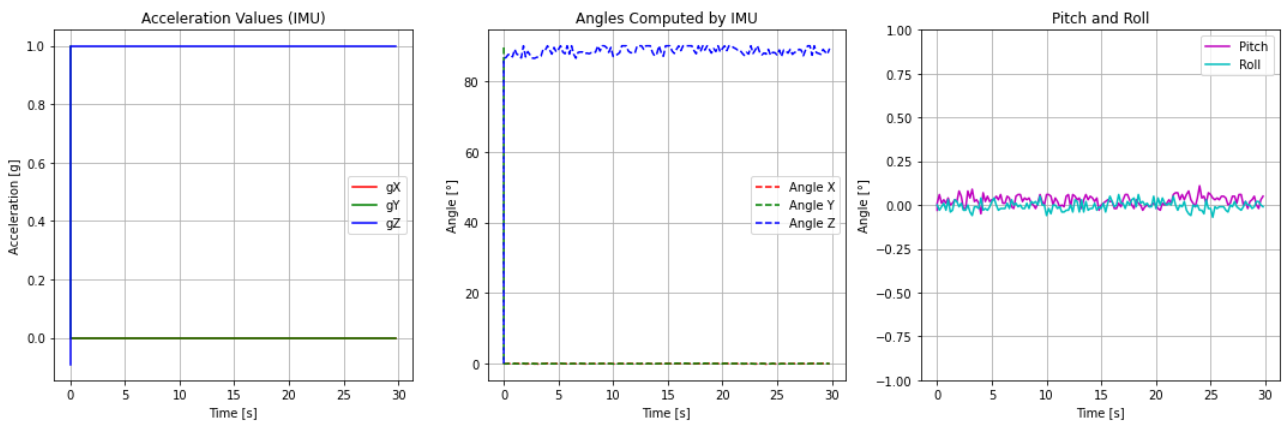

Figure 9. Static test results of the IMU at rest. Data show the expected stable output, with Pitch and Roll angles at 0° and the 1g gravity acceleration aligned with the Z-axis, validating the sensor baseline.

The results, presented in Figure 9, show that the Pitch and Roll angles remained stable and oscillated very close to 0°, with minimal variation attributed to the sensor's inherent noise. At the same time, the accelerometer data confirmed that the 1g gravity vector acted exclusively along the Z-axis. This behavior is both ideal and expected, validating the sensor calibration and the correct operation of the DMP at rest.

4.2 Test 2: Controlled Dynamic Validation

The second test aimed to validate the sensor's ability to accurately and consistently track large-amplitude rotations. The IMU was subjected to controlled manual movements, being rotated to angles of approximately ±90° around the Pitch and Roll axes independently.

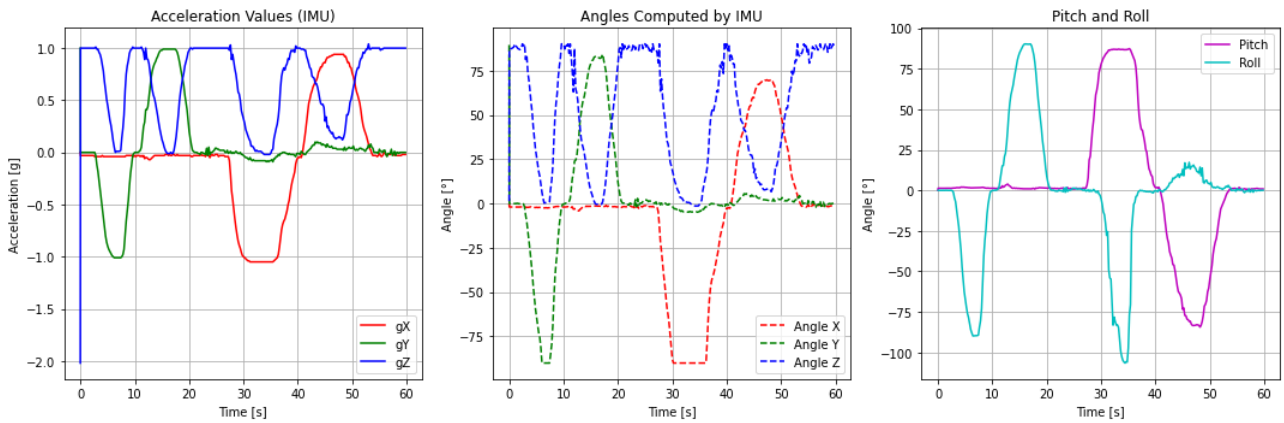

Figure 10. Controlled dynamic validation of the IMU (manual $\pm 90°$ rotations). Plots (left to right): gravity vector transfer, Euler angles, and Pitch/Roll tracking. Yaw instability (Gimbal Lock) is visible near $\pm 90°$ Pitch.

As illustrated in Figure 10, the Controlled Dynamic Pitch and Roll plot shows that the sensor was able to follow the angular trajectories clearly and continuously. The internal consistency of these measurements is corroborated by the accelerometer data in the Acceleration plot. It was observed that, when reaching 90° rotations, the 1g gravity vector was correctly transferred from the Z-axis to the corresponding horizontal axis (Y-axis for Roll rotations and X-axis for Pitch rotations), confirming that the orientation estimation is physically coherent.

This test also revealed an important artifact related to the use of Euler angles: the phenomenon known as Gimbal Lock. As shown in the Computed Angles plot, when the Pitch angle approached ±90°, the Yaw angle (Angle Z) became unstable. This well-known mathematical limitation can compromise control if the robot reaches large tilt angles, and its observation is essential to define safe operational limits for the control system.

4.3 Test 3: Characterization under Adverse Dynamic Conditions

Finally, a stress test was conducted to characterize the IMU behavior under conditions simulating external disturbances, such as shocks or vibrations. The sensor was subjected to rapid and random manual shaking.

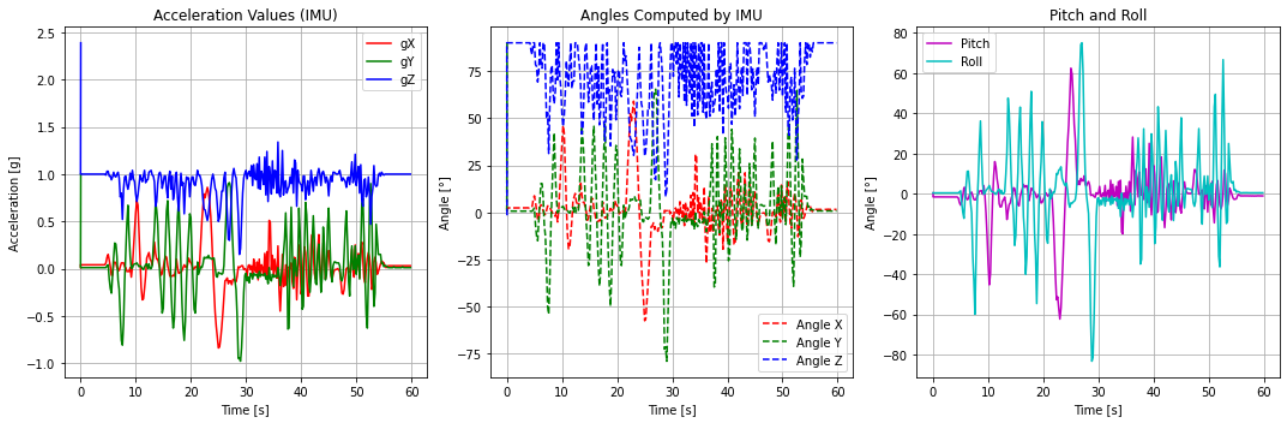

Figure 11. IMU performance under dynamic stress (random shaking test). Plots (left to right): raw acceleration, computed Euler angles, and Pitch/Roll angles. Significant linear accelerations ($> 1g$) correlate with increased noise and instability in orientation estimation.

The most illustrative result of this test comes from the accelerometer data. Unlike the previous tests, the acceleration values frequently exceeded 1g, indicating the presence of significant linear accelerations superimposed on the gravitational vector.

This "contamination" of the acceleration directly affects the DMP's data fusion algorithm, which relies on the gravity vector as a reference for correcting the gyroscope. Consequently, the orientation estimation, shown in Figure 11 "Pitch

and Roll" plot, becomes noticeably noisier and transiently less accurate during the shaking. This robustness analysis is crucial, as it demonstrates that the TWIP control system must be designed to withstand temporary degradations in sensor data quality during intense dynamic events.

5. CONCLUSIONS AND FUTURE WORK

This work presented the development of a functional two-wheeled inverted pendulum (TWIP) prototype, including mechanical design, electronic integration, and the foundation for a balance controller. A key focus was the experimental characterization of the ICM-20948 IMU, which is critical for reliable orientation estimation.

Static tests confirmed the sensor's baseline accuracy, controlled dynamic tests validated its ability to track large-angle rotations, and adverse dynamic tests highlighted performance degradation under high linear accelerations. These results provide essential insights for designing a robust balance control system and establishing operational limits, such as Gimbal Lock at ±90° Pitch.

The modular platform, built with accessible components, offers a reproducible foundation for embedded control studies of unstable systems, with potential for teaching and research in robotics and control. Future work includes implementing the balance control loop on the embedded system, conducting practical balancing tests, exploring more robust orientation representations (e.g., Quaternions), and developing filtering techniques to mitigate sensor noise under dynamic disturbances.

These steps aim to evolve the prototype into a complete and reliable platform for advanced research in multivariable control, state estimation, and autonomous robotics.

6. REFERENCES

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7. RESPONSIBILITY NOTICE

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